Yamini Mandadi

Software Engineer | Backend & Cloud Infrastructure | AI/ML Integration | Database Automation

Los Angeles, CA | yaminiofficial789281@gmail.com | linkedin.com/in/yaminimandadi | github.com/Yaminiiii7 | yaminiiii7.github.io

SUMMARY

Cloud/DevOps Engineer with 4+ years of professional experience in Microsoft Azure infrastructure, Terraform-based Infrastructure as Code, CI/CD automation, cloud security, and production reliability. Experienced in building reusable Terraform modules, automating deployments with Jenkins and GitHub Actions, implementing Azure RBAC and Key Vault security controls, and monitoring production environments using Azure Monitor, Log Analytics, and Application Insights. Skilled in Python, PowerShell, Bash, Docker, Kubernetes, and incident troubleshooting, with a focus on secure, repeatable deployments, high availability, and operational efficiency.

EDUCATION

Master of Science in Computer Science | *California State University, Los Angeles* — Los Angeles, CA

Bachelor of Technology in Computer Science and Engineering | *Reva University* — Bengaluru, India

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (Node.js, Express.js, React frontend), SQL, Bash/Shell Scripting, PowerShell, C++, C;

AI/ML & LLMs: LLMs (Llama 3.2), AI APIs, RAG, LangChain, Prompt Design, LLM Evaluation, AI Safety Guardrails, Vector Search (ChromaDB), Machine Learning, Hugging Face Transformers, PyTorch, Fine-Tuning

Backend & Distributed Systems: RESTful APIs, Microservices, Event Streaming (Kafka, Azure Event Hubs), Distributed Key Value Store (Redis), Schema-less Data Stores (MongoDB), Authentication & Authorization (JWT, OAuth 2.0, RBAC), Configuration Management

Databases: PostgreSQL, MySQL, Azure SQL/SQL Server, MongoDB, Elasticsearch/OpenSearch, Relational Databases, Data Modeling, Query Optimization, Backup/Restore, Replication, High Availability, Disaster Recovery

Cloud, Containers & CI/CD: Azure, AWS (ECR, IAM), VMware, Docker, Docker Compose, Kubernetes, Terraform, ARM Templates, Packer, Git (source control), GitHub Actions, Jenkins, Bitbucket, Linux Administration & Internals (Ubuntu, Kali Linux);

Observability & Reliability: Azure Monitor, Log Analytics (KQL), Application Insights, Prometheus, Grafana, Alertmanager, Distributed Logging, Tracing, Monitoring, Alerting, Performance Tuning, Root-Cause Analysis, Runbooks

Software Engineering Practices: Data Structures, Algorithms, Object-Oriented Design, Unit Testing (pytest, Jest), Test Automation, Code Reviews, Secure Coding (OWASP), Agile/Scrum, Version Control, Debugging, Documentation, Cross-Team Collaboration

Security & Tools: Microsoft Entra ID, Azure Key Vault, Azure Firewall, NSGs, Microsoft Defender for Cloud, Zero Trust, SSL/TLS, Nmap, Wireshark, Nessus, Metasploit, Jira, Confluence; AI Coding Tools: Claude Code, Cursor, Windsurf, GitHub Copilot

PROFESSIONAL EXPERIENCE

Cloud Platform Engineer | *Community Dreams Foundation* — Aug 2026 – Present

- Develop event-driven integrations with Azure Logic Apps and Azure Event Hubs (Kafka-compatible event streaming), applying integration patterns to build automated workflows, REST API integrations with web services, and data-ingestion pipelines across internal systems.
- Implemented a secure end-to-end CI/CD pipeline for Azure Kubernetes Service (AKS) using GitHub Actions and Argo CD, integrating code scanning, branch protection, Azure Workload Identity Federation and Helm-based deployments of containerized microservices from Azure Container Registry (ACR).
- Developed and orchestrated Azure Data Factory (ADF) pipelines to transfer product data between Azure SQL Database and Azure Blob Storage, configuring Linked Services, Datasets, Copy Data activities, and execution dependencies
- Automated AWS infrastructure EC2, S3, VPC, IAM, security groups, least-privilege access and network security controls using Terraform and developed CI/CD pipelines with Jenkins and Git to streamline infrastructure deployments, build validation and release workflows.
- Integrated CloudWatch monitoring and Sumo Logic alerts with Python AWS Lambda to detect EC2 application issues, trigger automated remediation, and improve infrastructure reliability while reducing manual intervention

Data Analyst | *California State University, Los Angeles* — Aug 2025 – May 2026

- Analyzed customer purchasing patterns and buying behavior using Python in Jupyter Notebook to uncover trends in transactions, customer segments, and product demand.
- Cleaned, processed, and explored large customer datasets through data preprocessing and EDA to improve data quality and identify key drivers of sales and repeat purchases.
- Built insightful visualizations and business-focused analyses to highlight customer segments, product affinity, and purchasing trends that support marketing and inventory decisions.
- Derived actionable insights on demand forecasting, customer retention, and revenue opportunities, presenting findings in a clear, decision-support format for business strategy.

System Engineer | *Tata Consultancy Services* — Jul 2023 – Sep 2024

- Developed reusable Terraform modules with remote state and environment-specific configuration for Dev, QA/Staging, and Production, standardizing configuration management and enabling consistent, repeatable deployments.
- Built and maintained DevSecOps CI/CD pipelines with Jenkins, Git, and Azure CLI, integrating automated testing, secret scanning, Checkov IaC checks, Terraform validate/plan, and approval gates to enforce secure coding practices before production deployments.
- Administered production Azure SQL/SQL Server databases: automated database backups, long-term retention, and restore validation with PowerShell and Python scripts, configured geo-replication and failover groups for high availability, investigated performance problems and backup/restore failures, and owned disaster recovery planning aligned with RPO/RTO targets.
- Implemented observability with Azure Monitor, Log Analytics (KQL), Application Insights distributed tracing, Prometheus, and Grafana, defining metrics, alerts, and dashboards to troubleshoot bottlenecks, latency, service crashes, and production issues via root-cause analysis.
- Designed secure, resilient network architecture with hub-and-spoke VNets, subnet segmentation, VNet peering, UDRs, Azure Firewall, NSGs, Application Gateway, Load Balancer, Private Endpoints, and Private DNS, reducing public exposure of Azure PaaS services.

- Collaborated with distributed teams across time zones in Agile/Scrum sprints (sprint planning, design reviews, pair programming, pull-request code reviews), integrating Jira with Bitbucket to trace issues to Git branches, PRs, and deployments.
- Mentored new trainees and led knowledge-sharing sessions on Terraform, CI/CD, and troubleshooting best practices; authored runbooks and internal knowledge base articles for operations and support teams.

Assistant System Engineer | *Tata Consultancy Services* Jul 2022 – Jul 2023

- Reduced Azure infrastructure costs by approximately 50–66% through utilization and performance analysis, resource rightsizing, App Service Plan optimization, and rehosting strategies using Azure Cost Management.
- Migrated application workloads from Azure App Service Plans to the serverless Consumption Plan, running proof-of-concept testing to validate compatibility, scalability, latency, networking, security, and deployment requirements.
- Implemented authentication and authorization controls using Microsoft Entra ID, OAuth 2.0 app registrations, Conditional Access, MFA, PIM, RBAC, Managed Identities, and Service Principals to enforce Zero Trust least-privilege access.
- Managed Azure Key Vault secrets and the SSL/TLS certificate lifecycle, and strengthened governance with Microsoft Defender for Cloud, Azure Policy, and storage network restrictions to safeguard customer data and application secrets.
- Troubleshot connectivity and security issues across VNets, subnets, NSGs, Azure Firewall, peering, and routing for production workloads, escalating complex issues to senior engineers and documenting resolutions.
- Supported Azure Data Factory (ADF) ETL pipeline development and execution, enabling secure connectivity to on-premises and private-network data sources through Self-hosted Integration Runtime (SHIR); configured linked services, datasets, Copy Data activities, and pipeline dependencies to transfer data from SQL Server to Azure Blob Storage.

Assistant System Engineer Trainee | *Tata Consultancy Services* Jul 2021 – Jul 2022

- Automated provisioning of Linux and Windows Azure Virtual Machines with ARM Templates, Terraform, and Azure CLI, standardizing compute configuration across development and production environments.
- Handled customer issue triage and support tickets for ITIL incidents, service requests, and change requests in ServiceNow ITSM within SLA targets, identifying root causes of customer-reported issues and contributing fixes to the internal knowledge base.
- Administered Active Directory, DNS, Microsoft Entra ID, and AWS IAM for hybrid/multi-cloud identity, authentication, and authorization, applying RBAC, MFA, NSG, firewall
- Assessed Azure security posture and Defender for Cloud recommendations; supported remediation through Azure Policy and storage restrictions

PROJECTS

Log Analytics & Incident Automation Platform | *Node.js, Kafka, Elasticsearch, PostgreSQL* GitHub

- Engineered Node.js/Express microservices for distributed logging (ingestion, indexing, search, and real-time analytics) on Elasticsearch (OpenSearch-compatible), using Kafka event streaming to buffer bursts and decouple producers from indexers.
- Designed RESTful API endpoints to query, filter, and aggregate logs; applied query optimization (index mappings, bulk indexing, pagination) and used traffic capture and replay with k6 load tests to find bottlenecks and reduce p95 latency.
- Integrated alerting with ServiceNow through the REST Table API to auto-create incidents on error-rate spikes.

AI-Native RAG Assistant (LLM & Vector Search) | *Python, Llama 3.2, LangChain, ChromaDB, FastAPI, Docker* GitHub

- Built a Retrieval-Augmented Generation (RAG) application in Python using Llama 3.2, LangChain, and ChromaDB vector search for semantic document retrieval and context-aware answers through a Gradio UI and a FastAPI REST API.
- Applied prompt design (system prompts, few-shot examples, citation grounding) and wrote an evaluation framework scoring retrieval precision@k, faithfulness, and latency on a golden Q&A set; A/B tested prompt variants via feature flags with config hot reloading.
- Added AI safety guardrails (PII redaction, prompt-injection filtering, out-of-scope refusals) and a pluggable provider layer supporting local Llama models and OpenAI-compatible AI APIs.
- Containerized with Docker Compose and automated CI/CD deployment pipelines in GitHub Actions that run pytest unit tests and linting before publishing images to Amazon ECR for repeatable deployments.

RESTful Music Playlist API | *Node.js, Express.js, MongoDB,Redis, JWT* GitHub

- Developed a RESTful backend API with Node.js, Express.js, MongoDB, and Mongoose (schema-less data store) using a modular route, model, and middleware architecture for user authentication, track search, and playlists.
- Implemented authentication and authorization with bcrypt password hashing, JWT-protected routes, input validation, and Redis-backed rate limiting; integrated the Last.fm API with caching in a Redis key-value store, retries, and a circuit breaker for resilience.
- Wrote unit and integration tests with Jest and Supertest and documented endpoints with OpenAPI/Swagger.

Fine-Tuned Text Classification Model | *Python, PyTorch, Hugging Face Transformers* GitHub

- Fine-tuned a transformer-based text-classification model to categorize online community posts by content and intent, enabling more efficient content discovery and identification of recurring discussion topics.
- Trained and evaluated the fine-tuned model against a baseline on labeled datasets, comparing precision, recall, and F1-score.

Network Scanning & Exploitation | *Linux, Python, Bash, Nmap, Nessus, Wireshark, Metasploit* GitHub

- Automated network security assessments on Kali Linux with Python and Bash scripts orchestrating Nmap, Nessus, Wireshark, and Metasploit for TCP/UDP port scanning, service enumeration, OS fingerprinting, and vulnerability validation in isolated test environments.
- Prioritized findings with CVSS-based risk scoring and wrote structured remediation reports.

CERTIFICATIONS & ACHIEVEMENTS

- Microsoft Certified: Azure Administrator Associate (AZ-104)
- Microsoft Certified: Azure Security Engineer Associate (AZ-500)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- Azure Security Hackathon Winner
- National Cyber League (NCL), Fall 2024: Team Game ranked 574th of 4,893 nationally (89th percentile; 93rd percentile in Log Analysis); Individual Game ranked 1,392nd of 8,484 (84th percentile; 94th percentile in Network Traffic Analysis)